

Apache Hadoop – Simple Practice Guide

Apache Hadoop is a **distributed data processing framework**.

1. What is installed

For learning, we run it in **pseudo-distributed mode**, where **one machine plays all roles**:

- **Client**: submits jobs and commands
- **NameNode**: manages file system metadata
- **DataNode**: stores file blocks
- **ResourceManager**: manages resources (YARN)
- **NodeManager**: executes tasks

This can be done on:

- **Linux**, or
- **Windows using WSL (Windows Subsystem for Linux)**

Reference installation : <https://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-common/SingleCluster.html>

2. How to start Hadoop (from Windows)

Step 1 — Open Linux

start the Linux environment inside Windows.

wsl

Step 2 — Start HDFS

starts NameNode, DataNode, and SecondaryNameNode.

start-dfs.sh

Step 3 — Start YARN

starts ResourceManager and NodeManager.

start-yarn.sh

Step 4 — Verify Hadoop is running

shows Hadoop Java processes.

jps

You should see: NameNode, DataNode, ResourceManager, NodeManager.

3. Basic HDFS commands

List HDFS root: shows directories in HDFS.

hdfs dfs -ls /

Create a directory in HDFS

hdfs dfs -mkdir /data

List the new directory

hdfs dfs -ls /data

4. Create a big file

Create a large file so HDFS must split it into blocks.

```
yes HADOOP | head -n 5000000 > bigfile.txt
```

Upload the file to HDFS: copies the file from Linux into HDFS.

```
hdfs dfs -put bigfile.txt /data
```

Verify the file in HDFS: shows the file stored in HDFS.

```
hdfs dfs -ls /data
```

5. Check how HDFS split the file into blocks

Show block information: displays how many blocks HDFS created.

```
hdfs fsck /data/bigfile.txt -blocks
```

Show block locations shows which DataNode stores each block.

```
hdfs fsck /data/bigfile.txt -blocks -locations
```

Note:

- In this setup, all blocks are on **one DataNode** because there is **one machine**

6. Simple MapReduce example (Word Count)

Run MapReduce WordCount: counts how many times each word appears in the file.

```
hadoop jar /usr/local/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.6.jar wordcount /data/bigfile.txt /out
```

7. How to stop Hadoop

Stop YARN

```
stop-yarn.sh
```

Stop HDFS

```
stop-dfs.sh
```

Exit Linux

```
exit
```